

Section : Physics

Q.1 During transmission of electrical energy by transformer, near the consumer

1. the voltage is stepped up
2. the voltage is stepped down
3. the voltage and current both are stepped up
4. the voltage and current both are stepped down

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Monochromatic light of wavelength 600 nm is incident from air onto a water surface.

Find the wavelength and speed of light respectively in water. (Given: Refractive index of water $n = 4/3$)

1. 450 nm; 3×10^8 m/s
2. 450 nm; 2.25×10^8 m/s
3. 800 nm; 3×10^8 m/s
4. 800 nm; 2.25×10^8 m/s

Options 1. 1

2. 2
3. 3
4. 4

Q.3 In Young's double slit experiment, two slits $d = 0.12$ mm apart are illuminated by monochromatic light of wavelength 480 nm. The screen is 1.0 m away from the slits. The distance of the second bright fringe from the central maximum will be

1. 8×10^{-4} m
2. 4×10^{-4} m
3. 5×10^{-3} m
4. 8×10^{-3} m

Options 1. 1

2. 2
3. 3
4. 4

Q.4 A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 1 A. If the number of turns is 800 per metre, the magnetic field inside solenoid will be

1. 0.40 T
2. 4.02 T
3. 40.2 T
4. 0.04 T

Options 1. 1

2. 2
3. 3
4. 4

Q.5 A battery of emf 3 V and internal resistance r is connected in series with a resistance of $R = 55 \Omega$ through an ammeter of resistance $R_A = 1 \Omega$. The ammeter reads a current $I = 50$ mA. Find the internal resistance r of the battery.

1. 2Ω
2. 3Ω
3. 4Ω
4. 5Ω

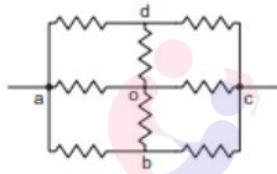
Options 1. 1

2. 2
3. 3
4. 4

Q.6

The equivalent resistance between the points a and c of the network shown in the figure, would be

(Here each resistance is equal to r .)



1. $3r/2$
2. $2r/3$
3. $r/3$
4. $r/2$

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which of the following equations represents the Ampere-Maxwell law?

1. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$
2. $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\epsilon_0}$
3. $\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi_B}{dt}$
4. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The time required for a 50 Hz alternating current to reach the peak value starting from zero

1. 100 s
2. $1/200$ s
3. 200 s
4. 50 s

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Two parallel large thin metal sheets have equal charge densities of $5.31 \times 10^{-11} \text{ C/m}^2$ of opposite signs.

If the permittivity of free space is $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$,

the electric field between these sheets will be:

1. 60 N/C
2. 47 N/C
3. 6 N/C
4. 4.7 N/C

Options 1. 1

2. 2
3. 3
4. 4

Q.10 When light travels from one medium to another, then its

1. frequency changes but wavelength remains the same
2. wavelength changes but frequency remains the same
3. both frequency and wavelength remain unchanged
4. both frequency and wavelength change

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Choose the correct statement(s) from the following:

(A) With increasing reverse bias voltage, the barrier potential across the junction of p-n diode decreases.

(B) A p-type semiconductor can be obtained by doping Si or Ge with trivalent atoms.

(C) A rectifier rectifies dc voltages.

(D) Pure semiconductors are called extrinsic semiconductors.

Choose the *correct* answer from the options given below:

1. (A) only
2. (B) only
3. (C) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.12 The magnetic flux (Φ) through the area of cross-section of a current-carrying coil is proportional to the current (i) i.e.,

$$\Phi = Li$$

(A) L is self-inductance of the coil

(B) The self-inductance of the coil does not depend on its geometry

(C) self-inductance is the electromagnetic analogue of mass in mechanics

(D) The self-inductance of the coil depends on the permeability of the medium inside coil

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (B), (C) and (D) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.13 In a coil, a current of 0.2 A increases at the rate of 0.4 A s^{-1}

If the power flow in the coil is 0.6 J s^{-1} the inductance of the coil is:

1. 7.5 H
2. 3 H
3. 1.2 H
4. 0.75 H

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Match List-I with List-II

List-I	List-II
Name of the Particle	Charge and Mass
(A) Proton	(I) $+2e$; 4u
(B) Neutron	(II) $+e$; 1u
(C) Deuteron	(III) 0; 1u
(D) α particle	(IV) $+e$; 2u

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 An oil drop of mass m carrying a charge +q is held stationary in air by an electric field E.

The electric field is correctly given by

1. $E = mg/q$ acting upwards
2. $E = mg/q$ acting downwards
3. $E = q/mg$ acting upwards
4. $E = q/mg$ acting downwards

Options 1. 1

2. 2
3. 3
4. 4

Q.16 E and V, respectively, denote electric field intensity and electrostatic potential at a point. The correct relation between the two is

1. $E = dV/dr$
2. $E = -dV/dr$

3. $V = dE/dr$
4. $V = -dE/dr$

Options 1. 1

2. 2
3. 3
4. 4

Q.17 A 200-turn coil of self-inductance 20 mH carries a current of 4 mA.
The magnetic flux linked with each turn of the coil is:

1. 16×10^{-3} Wb
2. 8×10^{-5} Wb
3. 4×10^{-7} Wb
4. 2×10^{-6} Wb

Options 1. 1

2. 2
3. 3
4. 4

Q.18 An electron of mass 9.11×10^{-31} kg is moving with velocity 5.4×10^6 m/s.

Its momentum will be

1. 4.92×10^{-24} kg m/s
2. 4.66×10^{-25} kg m/s
3. 6×10^{-37} kg m/s
4. 6.6×10^{-22} kg m/s

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Two circular coils of diameter 10 cm and 20 cm, carrying equal currents have the same number of turns. The ratio of magnetic fields produced at the center of coils, respectively will be

1. 3:2
2. 2:3
3. 1:2
4. 2:1

Options 1. 1

2. 2
3. 3
4. 4

Q.20 A point source of light is placed at a distance twice the focal length of a converging lens. If the focal length of the lens is f , the intensity of the image of light source on the other side of the lens will be maximum at

1. f
2. $2f$
3. between f and $2f$
4. $> 2f$

Options 1. 1

2. 2
3. 3
4. 4

Q.21 The ratio of densities of nuclei with mass numbers A and $2A$, respectively is

1. 1:2
2. 2:1
3. 1:1
4. $1:2^{\frac{1}{3}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.22 Which of the following is not a unit of electric field intensity?

1. $N C^{-1}$
2. $N A^{-1} s^{-1}$
3. $V m^{-1}$

4. V m

Options 1. 1

2. 2
3. 3
4. 4

Q.23 The materials and devices do not obey Ohm's law when
(A) The relation between V and I is not unique, i.e., there is more than one value of V for the same current I

- (B) The relation between V and I depends on the sign of V
- (C) V ceases to be proportional to I
- (D) The relation between V and I is linear

Choose the *correct* answer from the options given below:

1. (B), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The atomic numbers and mass numbers of the nuclei of hydrogen, deuterium and tritium, respectively are in the ratio

1. 1:2:3 and 1:2:3
2. 3:2:1 and 1:2:3
3. 1:1:1 and 1:2:3
4. 1:2:3 and 1:1:1

Options 1. 1

2. 2
3. 3
4. 4

Q.25 Choose the phenomenon from the following, which shows that light is a transverse wave.

1. Diffraction
2. Interference
3. Reflection
4. Polarisation

Options 1. 1

2. 2
3. 3
4. 4

Q.26 An a.c. voltage source of fixed amplitude E_0 and variable angular frequency ω is connected with an electric bulb of resistance R (inductance zero) and a capacitor of capacitance C in series.

On increasing the angular frequency of the source, the total impedance of the circuit

1. increases and the bulb glows brighter
2. decreases and the bulb glows dimmer
3. remains the same and the bulb glows brighter
4. decreases and the bulb glows brighter

Options 1. 1

2. 2
3. 3
4. 4

Q.27 A slab of material of dielectric constant K has the same area as the plates of a parallel-plate capacitor but has a thickness $\frac{1}{4}d$, where d is the separation between the plates.

If the slab is inserted between the plates, the potential difference becomes:

(Given: Potential difference across the capacitor without dielectric is V_0)

1. $V = V_0\left(\frac{3K+4}{4}\right)$
2. $V = V_0\left(\frac{3K+1}{4K}\right)$
3. $V = V_0\left(\frac{K+1}{4K}\right)$
4. $V = V_0\left(\frac{3K+4}{4K}\right)$

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Photoelectric current depends on

- (A) the intensity of incident light
(B) the potential difference applied between the two electrodes
(C) the nature of the emitter material
(D) the conductivity of the emitter material
Choose the correct answer from the options given below:
1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.29 A bar magnet with pole strength m and magnetic dipole moment p is cut into two identical pieces along its length.

Then for each of the two pieces

1. pole strength and dipole moment are m and p , respectively
2. pole strength and dipole moment are $m/2$ and $p/2$, respectively
3. pole strength is $m/2$ and dipole moment is p
4. pole strength is m and dipole moment is $p/2$

Options 1. 1

2. 2
3. 3
4. 4

Q.30 A total of 5.0×10^{16} electrons per second go through a conductor of area 2 mm^2 .

The current density will be:

1. 4000 A m^{-2}
2. $4 \times 10^3 \text{ A m}^{-2}$
3. 4 A m^{-2}
4. 8 A m^{-2}

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Two wires of equal lengths carrying equal currents bent in the form of two loops, one in square shape and the other in circular shape having one turn only are suspended in same, uniform magnetic field perpendicular to the plane of loops.

Choose the correct option

1. both the loops experience equal torque
2. both the loops experience zero torque
3. Square loop experiences smaller torque
4. Circular loop experiences the smaller torque

Options 1. 1

2. 2
3. 3
4. 4

Q.32 The colour of light which is scattered the most by the atmospheric molecules is

1. Red
2. Green
3. Blue
4. Yellow

Options 1. 1

2. 2
3. 3
4. 4

Q.33 A 50 Hz alternating current is flowing in a 28 mH coil. The reactance of the coil is

1. 88Ω
2. 4.4Ω
3. 8.8Ω
4. 0.88Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.34 A thin lens L_1 of focal length 20 cm placed in combination with another thin lens L_2 acts as a lens of focal length 12.5 cm. The power of L_2 is

1. 3 D
2. 8 D

3.5D
4.1D

- Options 1. 1
2. 2
3. 3
4. 4

Q.35 Match List-I with List-II

List-I (Physical Quantity)	List-II (Unit)
(A) Magnetic flux	(I) Tm A^{-1}
(B) Magnetic permeability	(II) Am^2
(C) Magnetisation	(III) Tm^2
(D) Magnetic moment	(IV) Am^{-1}

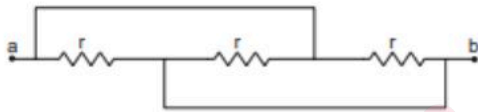
Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

- Options 1. 1
2. 2
3. 3
4. 4

Q.36

The equivalent resistance of the network shown in figure between the points **a** and **b** will be



1. $3r$
2. $r/3$
3. r
4. $3r/2$

- Options 1. 1
2. 2
3. 3
4. 4

Q.37 A charged particle enters a magnetic field with velocity v at an angle θ with the field and experiences a magnetic force.

The kinetic energy of the particle

1. becomes maximum if $\theta = 90^\circ$
2. becomes maximum if $\theta = 0^\circ$
3. becomes minimum if $\theta = 180^\circ$
4. is not affected by magnetic force

- Options 1. 1
2. 2
3. 3
4. 4

Q.38 Identify the correct statements from the following for an electrical circuit.

- (A) An ammeter of very low resistance is connected in series in the circuit.
 - (B) A Voltmeter of high resistance is connected in series with a resistance in the circuit.
 - (C) A current I coming at a point P of a circuit, divides into 4 small currents flowing through different branches and all the currents again combine to give current I at the next point S of the circuit. The branches are in parallel.
 - (D) The internal resistance of the batteries can always be neglected
- Choose the *correct* answer from the options given below:
1. (A) and (B) only
 2. (A) and (C) only
 3. (A) and (D) only
 4. (B) and (D) only

- Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.39 In an intrinsic semiconductor, the relation between the number of holes (n_h) and conduction electrons (n_e) will be:

- 1. $n_h = n_e$
- 2. $n_h > n_e$
- 3. $n_h = 2n_e$
- 4. $n_h < n_e$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 Accelerated charges radiate electromagnetic waves. Choose the correct key features of electromagnetic waves from the following:

- (A) The electric field is always parallel to the magnetic field.
- (B) The electric and magnetic fields are always normal to the direction in which the wave is travelling.
- (C) The electric and magnetic fields sometimes vary sinusoidally.
- (D) The electric and magnetic fields vary with the same frequency.

Choose the correct answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (B) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 The electron in a hydrogen atom emits wavelengths $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ in the following transitions (A), (B), (C), and (D), respectively. Arrange the transitions in the increasing order of emitted wavelengths:

- (A) $n = 3$ to $n = 2$
- (B) $n = 4$ to $n = 3$
- (C) $n = 4$ to $n = 2$
- (D) $n = 2$ to $n = 1$

Choose the correct answer from the options below:

- 1. (A), (B), (C), (D)
- 2. (A), (C), (B), (D)
- 3. (D), (A), (B), (C)
- 4. (D), (C), (A), (B)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.42 A convex lens has 20 cm focal length in air. Its focal length in water would be (Given : Refractive index of water = 1.33, refractive index for glass = 1.5)

- 1. + 78.2 cm
- 2. + 27.8 cm
- 3. - 82.7 cm
- 4. - 78.2 cm

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.43 The wavelength range of X-rays is:

- 1. 700nm to 400 nm
- 2. 1 nm to 10^{-3} nm
- 3. $< 10^{-3}$ nm
- 4. 0.1 m to 1 mm

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

- Q.44 A proton and a deuteron execute circular orbits having same radii of 0.5 m in a plane perpendicular to a uniform magnetic field B. If the kinetic energy of the deuteron is 50 keV, the K.E. of the proton will be
1. 200 keV
 2. 100 keV
 3. 50 keV
 4. 25 keV

Options 1. 1
2. 2
3. 3
4. 4

- Q.45 The electric potential at a point in an electric field is 8×10^4 V. The work done to bring a point charge 5×10^{-9} C from infinity to this point is:
1. 0.5×10^{-3} J
 2. 4×10^{-4} J
 3. 6×10^{-4} J
 4. 4×10^{-6} J

Options 1. 1
2. 2
3. 3
4. 4

- Q.46 An electron accelerated with potential V has a de-Broglie wavelength λ associated with it. If the potential is changed to 4 V, the wavelength becomes
1. $\lambda/4$
 2. $\lambda/2$
 3. λ
 4. 2λ

Options 1. 1
2. 2
3. 3
4. 4

- Q.47 The ratio of equivalent capacitance in series and parallel combinations of two capacitors of capacity 5 F and 20 F will be
1. 1:5
 2. 4:25
 3. 4: 1
 4. 25:4

Options 1. 1
2. 2
3. 3
4. 4

- Q.48 E_{Ge} , E_{Si} , and E_C denote the ionization energy values for Ge, Si and C, respectively. The values correctly arranged in increasing order are:
1. E_C , E_{Ge} , E_{Si}
 2. E_{Ge} , E_C , E_{Si}
 3. E_{Si} , E_{Ge} , E_C
 4. E_{Ge} , E_{Si} , E_C

Options 1. 1
2. 2
3. 3
4. 4

- Q.49 Which of the following statements are correct?
- (A) the mid-point between the two similar and equal point charges located nearby, is called neutral point.
- (B) There are maximum number of field lines at the neutral point.
- (C) The work done to bring any positive charge from infinity to a point in an electric field is called the electric potential at that point.
- (D) The work done to move a charge from one point to another point on the equipotential surface will be zero.
- Choose the correct answer from the options given below:
1. (A) and (D) only
 2. (A), (B) and (C) only
 3. (B) and (C) only
 4. (C) and (D) only

Options 1. 1
2. 2
3. 3

Q.50 The terminal voltage of a cell

1. is always equal to its emf
2. is always less than its emf
3. is always more than its emf
4. may be equal or less than or greater than its emf

Options 1. 1

2. 2

3. 3

4. 4